

Ferdaous Sarhani

AI & Data Science Engineering Student | Machine Learning | Full-Stack Development
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PROFESSIONAL SUMMARY

Highly motivated AI & Data Science Engineering student at TEK-UP University, with a Bachelor's degree in Business Computing from ESSEC. Proven track record of building end-to-end AI/ML systems, RAG-based agents, full-stack web applications, and data pipelines.

Ranked Top 10 of 126 teams in a national Kaggle data science competition and Top 32 of 392 teams across 14 countries in the Orange Digital Center Champions 2025. Seeking an internship or junior role in Data Engineering, Machine Learning, or AI Development.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, PHP, Java, HTML/CSS, SQL

AI / ML / DL: Supervised & Unsupervised Learning, Deep Learning, RNNs, Transformers, NLP, TensorFlow, PyTorch, scikit-learn, Keras, LangChain, RAG, ChromaDB, LLM integration

Web & Mobile: React, Node.js, Bootstrap, JavaFX, Android Studio (Android SDK), REST APIs, Streamlit

Databases: MySQL, MongoDB, NoSQL, Oracle, Firebase / Firestore

DevOps & Tools: Git, GitHub, GitLab, Odoo, Apache POI

Methodologies: Agile, SCRUM, CRISP-DM, MVC, UML

Languages: Arabic (Native), English (Fluent), French (Advanced)

PROJECTS

HR Intelligent Assistant — Autonomous AI Agent with RAG | [SarhaniFerdaous/HR-Intelligent-Assistant](#) 2025
Python · LangChain · ChromaDB · Groq / Llama 3.3 70B · Streamlit · HuggingFace Embeddings · REST APIs

- Built a multi-tool AI agent that answers HR questions exclusively from verified company documents (PDF, DOCX, TXT, CSV), eliminating hallucination via a RAG pipeline using ChromaDB and the all-MiniLM-L6-v2 embedding model.
- Integrated real-time API tools: public holiday lookup (date.nager.at) and currency conversion (open.er-api.com) for international salary queries, etc..
- Added compliance-checking and HR-contact routing tools, with transparent reasoning steps surfaced to the user via LangChain agent memory.
- Supports multilingual responses (English / French / Arabic and Tunisian Dialect), dynamically matching the employee's input language.

"Soulier d'Or" — AI-Powered Football Experience Platform | [pfe / SoulierDOr](#) · [GitLab](#) 2024 – 2025
Python · Machine Learning · Deep Learning · LangChain · Grafana · REST APIs · CRISP-DM · SCRUM | PFE Internship @ Tritux

- Designed and trained a vote-prediction model and a user-segmentation model, both deployed as independent REST APIs enabling real-time automated decision-making.
- Integrated model outputs to automatically trigger personalised push notifications, driving targeted user engagement through ML-driven content delivery.
- Built interactive Grafana dashboards to monitor model performance KPIs and user engagement metrics in real time.
- Applied the CRISP-DM data science lifecycle end-to-end (business understanding → data preparation → modelling → evaluation → deployment) within an Agile/SCRUM team.

InfoZone — Full-Stack E-Commerce Marketplace 2024
React · Bootstrap · Node.js · Firebase · Power BI · SEO

- Developed a full-stack marketplace covering product management, order processing, payments, and user authentication.

- Integrated Power BI dashboards for real-time sales analytics; implemented SEO optimisation and bilingual (Arabic/French) support.

Essectify — Teacher Absence Management Android App
2023 – 2024

Java · Android Studio · Firestore Firebase · Apache POI

- Built a mobile application managing teacher schedules and absences, with automated Excel report generation via Apache POI and Firebase push notifications.

Quiz Generator Platform | github.com/SarhaniFerdaous/QuizGenerator
2023

Java · XML · Maven

- Architected a full quiz management system supporting quiz creation, editing, and deletion with a role-based access control model (admin / teacher / student).
- Implemented result tracking and performance analytics per student, giving instructors clear visibility into class-level outcomes.
- Designed an intuitive XML-driven UI using JavaFX Scene Builder, ensuring a clean and responsive user experience without external CSS frameworks.
- Structured the project with Maven for dependency management and build automation, making it portable and easy to deploy across environments.

CVGEN — Personalised CV Generator
2022 – 2023

PHP OOP · PDO · MySQL · MVC

- Designed and developed a web application that allows users to create, customise, and export professional CVs in both PDF and HTML formats from a single interface.
- Implemented the MVC architecture pattern to cleanly separate business logic, data access (PDO/MySQL), and presentation layers, making the codebase maintainable and extensible.
- Secured all user data with parameterised queries via PDO to prevent SQL injection, and applied server-side input validation throughout the form pipeline.
- Built a live CV preview feature so users can see formatting changes in real time before exporting, reducing friction in the generation workflow.

EDUCATION

Engineering Cycle — AI & Data Science (in progress)
2025 – present

TEK-UP University · Tunis, Tunisia

Bachelor's Degree in Business Computing (Licence Informatique de Gestion)
2023 – 2025

ESSEC · Tunis, Monfleury, Tunisia · PFE completed at Tritux (company internship)

Scientific Baccalaureate
2020 – 2021

Lycée Taha Hussein · Mégrine, Ben Arous, Tunisia

COMPETITIONS & HACKATHONS

IEEEExtreme 19.0 — 24-Hour Global Programming Contest
2025

- Ranked 5th among all TEK-UP participants in the annual IEEE-organised 24-hour competitive programming competition.

Orange Digital Center Champions 2025 — CodinGame Optimisation Contest
2025

- Ranked 32nd out of 392 teams across 14 countries (1,354 participants) in an 8-hour code-optimisation marathon on CodinGame.

Sesame x JCI — Stroke Prediction Kaggle Challenge
2025

[Sesame-JCI-Stroke-Prediction | Kaggle](#)

- Competed in a medical AI challenge focused on stroke prediction using real-world clinical datasets; applied classification and feature engineering techniques.

GO DATA SCIENCE 5.0 — National Kaggle Competition

2025

[GO DATA SCIENCE 5.0 | Kaggle](#)

- Ranked 9th out of 126 competing groups in a national data science challenge hosted on Kaggle.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

IEEE TEK-UP Student Branch — Vice Chair Technical Activities (VCTA)

2025 – Present

- Leading technical workshops, study groups, and coding sessions for the TEK-UP IEEE student Branch.

IBI 2025 (2nd Conference on Intelligence in Business & Industry) — Organising Committee Member

April 2025

- Contributed to organising a two-day international academic conference held at ALECSO, Tunis.

infoLAB ESSECT — Human Resources Assistant

2024 – 2025

- Coordinated recruitment, onboarding, and internal communications for a student technology lab.

ENACTUS ESSECT — Member

2023 – 2024

- Contributed to social entrepreneurship projects, strengthening project management and teamwork skills.

AIESEC TACAPES — Marketing Manager & Talent Management Manager

2021 – 2023

- Marketing & Information Management: Built data-driven marketing campaigns and developed data processing skills using Google Sheets and reporting tools.
- Talent Management & Exchange: Led full recruitment cycles and established a reputation as an effective communicator and team leader.

WORKSHOPS & TRAINING

- NLP for Business — Workshop, IBI 2025 Conference, ALECSO Tunisia (April 2025)
- UI/UX Design & Design Thinking — ENACTSPARK Seminar, Enactus ESSECT (December 2023)
- Public Speaking & Debate — Interactive Training Day, Mounathara Initiative (2022)

Certifications in progress: PCAP – Python Certified Associate Programmer, AWS AI Practitioner